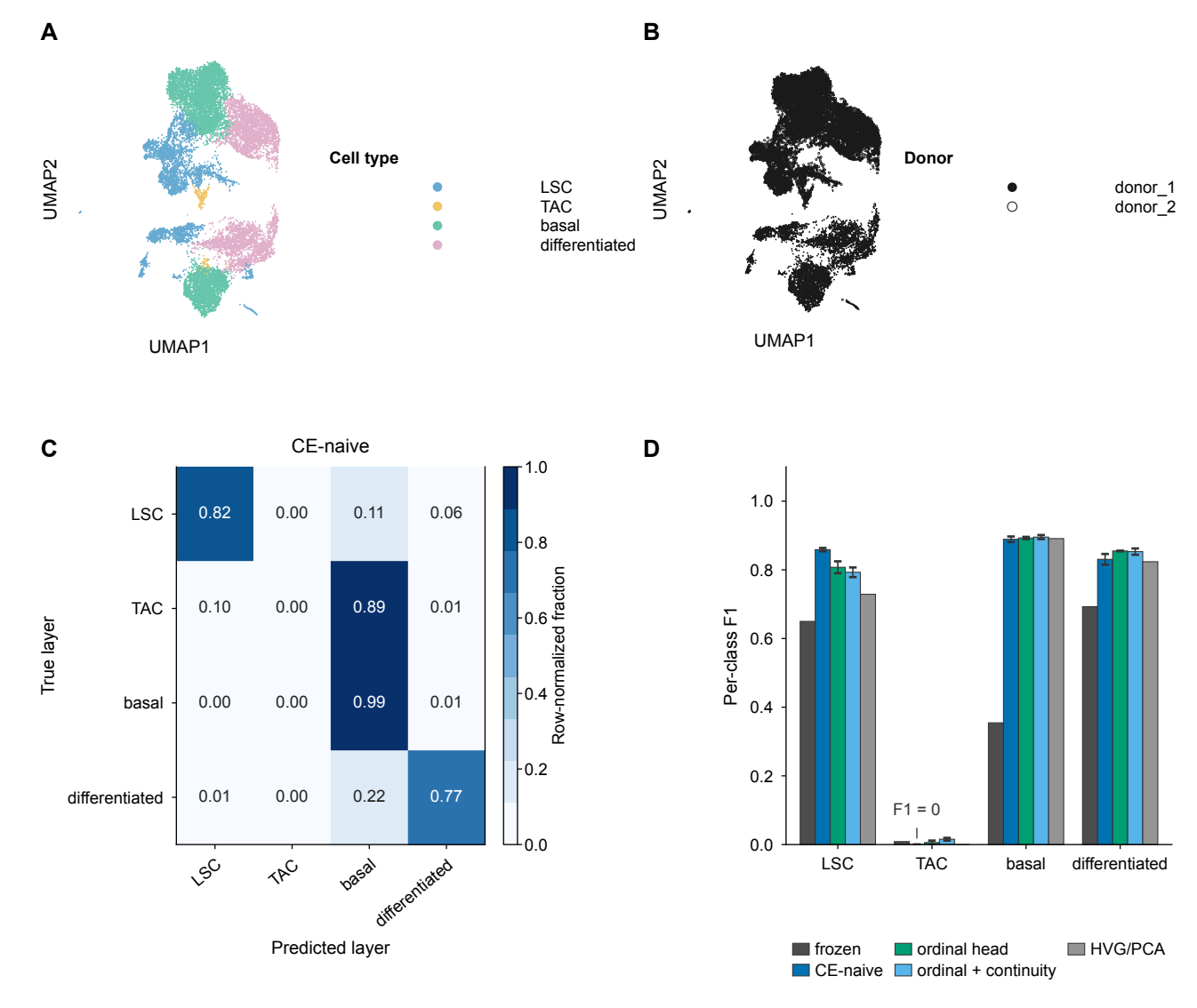

A
UMAP2
UMAP1
Cell type
LSC
TAC
basal
differentiated
B
UMAP2
UMAP1
Donor
donor_1
donor_2
C
CE-naive
True layer
LSC
TAC
basal
differentiated
0.82
0.00
0.11
0.06
0.10
0.00
0.89
0.01
0.00
0.00
0.99
0.01
0.01
0.00
0.22
0.77
Row-normalized fraction
1.0
0.8
0.6
0.4
0.2
0.0
Predicted layer
D
Per-class F1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0
LSC
TAC
basal
differentiated
frozen
CE-naive
ordinal head
ordinal + continuity
HVG/PCA